ECE 647: Lecture 17

Geometric Programming

Commonly used in Communications and Networking for applications such as Power Control and Networking Congestion Control.

Monomial
It's called a monomial because its a single term polynomial. The single term can have multiple variables with different powers (i.e. there are n unique x variables shown below).

$$f(x) = c \cdot x_1^{a_1} \cdot x_2^{a_2} ... x_n^{a_n}$$
$$c > 0, a_i \in \mathcal{R}$$

Posynomial
A posynomial ("positive polynomial") is formed by the sum on monomials.

$$f(x) = \sum_{k=1}^{K} c_k \cdot x_1^{a_{1k}} \cdot x_2^{a_{2k}} ... \cdot x_n^{a_{nk}}$$
$$c_k > 0 \ \forall k$$

Geometric Programming Form
Where $f_0, ..f_m$ are **posynomials** and $h_1, ...h_p$ are **monomials**

$$\begin{aligned} &\min f_0(x) \\ &\text{s.t. } f_i(x) \leq 1;\ i = 1, ..m \\ &\quad\ h_i(x) = 1;\ i = 1, ..p \\ &\quad\ x > 0 \end{aligned}$$

Examples
1

$$
\begin{aligned}
&\max x^{-1}y \\
&\text{s.t. } 2 < x < 3 \\
&\quad x^2 + \frac{3y}{z} \leq \sqrt{y} \\
&\quad \frac{x}{y} = z^2
\end{aligned}
$$

We can convert this into Geometric Programming form with algebraic manipulations. Convert the objective function to a minimum, convert inequality constraints to be posynomials ¡1, and convert equality constraints to monomials ¡1.

$$
\begin{aligned}
&\min x^{-1}y \\
&\text{s.t. } 2x^{-1} < 1 \\
&\quad \frac{x}{3} < 1 \\
&\quad x^2y^{-0.5} + 3y^{0.5}z^{-1} \leq 1 \\
&\quad xy^{-1}z^{-2} = 1
\end{aligned}
$$

2

Convert Geometric Programming Form to Convext.

- $x_i = e^{y_i}$
- $c_{i,k} = e^{b_{i,k}}$

$$
\begin{aligned}
&\min \sum_{k=1}^{K_0} e^{a_{0,k}^T y + b_0 k} \\
&\text{s.t. } \sum_{k=1}^{K_i} e^{a_{i,k}^T y + b_i k} < 1 \\
&\quad e^{g_i^T y + h_i} = 1
\end{aligned}
$$

Now we take a log, which preserves convexity. This makes the problem a "log sum exponential" which is convex.

$$
\begin{aligned}
\min \log(\sum_{k=1}^{K_0} e^{a_{0,k}^T y + b_0 k}) &\rightarrow \text{Convex Function} \\
\text{s.t.} \log(\sum_{k=1}^{K_i} e^{a_{i,k}^T y + b_i k}) < 0 &\rightarrow \text{Convex Set} \\
\log(e^{g_i^T y + h_i}) = 0 &\rightarrow \text{Linear}
\end{aligned}
$$

Thus posynomials $\Leftrightarrow$ convex functions and monomials $\Leftrightarrow$ linear functions.

Linear Fractional Programming

$$
\begin{aligned}
\min \frac{c^T x + d}{e^T x + f} \\
\text{s.t. } e^T x + f > 0 \\
Gx < h \\
Ax = b
\end{aligned}
$$

- We want a one-to-one mapping x $x \leftrightarrow (y, x), z > 0$
- y is a vector, z is a scaler
- $y = \frac{x}{e^T x + f}$
- $z = \frac{1}{e^T x + f} > 0$

For every x with $e^T y + f > 0$, then (y,z) can be obtained from the above equations.

Given (y,x) can we find x? This is required for a one-to-one mapping.

- Given (y,x) we can find $x = \frac{y}{z}$
- $z = \frac{1}{e^T (\frac{y}{z}) + f}$

- $e^T y + fz = 1$

Thus a one-to-one mapping exists.

$$
\begin{align}
&x; e^T x + f > 0 \tag{1}\\
&\text{to} \tag{2}\\
&(y, z); z > 0 \ \& \ e^T y + fz = 1 \tag{3}
\end{align}
$$

Thus this problem converts to

$$
\begin{aligned}
&\min c^T y + dz \rightarrow \text{linear!}\\
&\text{s.t. } z > 0 \rightarrow \text{linear!}\\
&\qquad e^T y + fz = 1 \rightarrow \text{linear!}\\
&\qquad Gy \leq hz \rightarrow \text{linear!}\\
&\qquad Ay \leq bz \rightarrow \text{linear!}
\end{aligned}
$$

Note that the (y,z) solution converts to an x via $(y^*, z^*) \rightarrow x^* = \frac{y^{*T}}{z^*}$

Note z $\geq$ 0 because at z goes to 0 as x goes to +inf (never goes negative).